Prevalence and rationale for covariate adjustment in cluster randomized trials: A scoping review

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Overview

Introduction & Background

- What are cluster randomized trials (cRCTs)?
- What is covariate adjustment?
- What is known in iRCTs vs cRCTs?
- Inspiration for this study
- Why LMICs?
- Why are we doing this review?

Procedures & Methods

- Data source
- Data extraction form
- Data extraction process

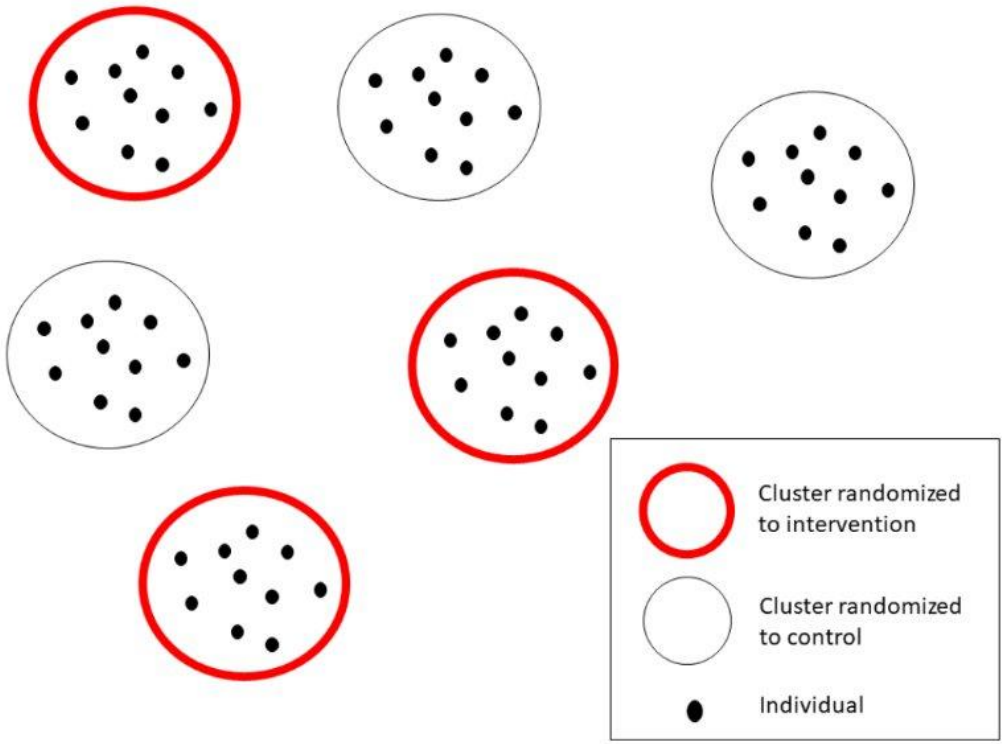
Results & Next Steps

- Results
- Next steps

Introduction & Background

What are cluster randomized trials (cRCTs)?

- Entire groups (clusters) randomized, not individuals
  - Examples: schools, clinics, villages
- Reduces contamination
- Often used in public health settings
- Fewer randomization units → covariate imbalance more likely



What is covariate adjustment?

- Adding baseline variables to statistical models estimating the treatment effect on outcomes
- Why?
  - Improves precision → increases statistical power
  - Adjusts for design variables
  - Handles chance covariate imbalances

Pocock et al. 2002

What is known in iRCTs vs. cRCTs?

What We Know (From iRCTs)	What We Don't Know (Gaps in cRCTs)
Existing guidance and literature on covariate adjustment	Less guidance specific to cRCTs
Explanations of when, why, and how to adjust for covariates	Limited systematic reviews examining covariate adjustment practices
Established best practices for choosing covariates and incorporating them into models	Lack of clarity of covariates being prespecified and adjusted for
Recommendations are frequently updated and adopted in the iRCT space	Last review was based on older trials

Inspiration for this study

- The Bachpan Study
  - Cluster randomized trial in Pakistan
  - Mental health intervention for mothers
  - Raised questions:
    - What covariates are adjusted for in low- and middle-income country (LMIC) cluster trials?
    - Do papers explain why they adjust?



Maselko et al. 2020

Why LMICs?

Many LMIC Trials	Variation in Methods	Gaps in Reporting
LMICs conduct a large number of community and public-health cRCTs	Methods and adjustment practices differ	Reporting on analysis decisions are inconsistent, making it unclear what was adjusted for

Goldstein et al. 2024

Why are we doing this review?

- Last cRCT covariate adjustment review was 10+ years ago
  - Used pre-2010 articles
  - No clear understanding of current covariate adjustment practices in cluster randomized trials
- Review of current practices can inform future work on guidance/recommendation documents

Wright et al. 2015

Procedures & Methods

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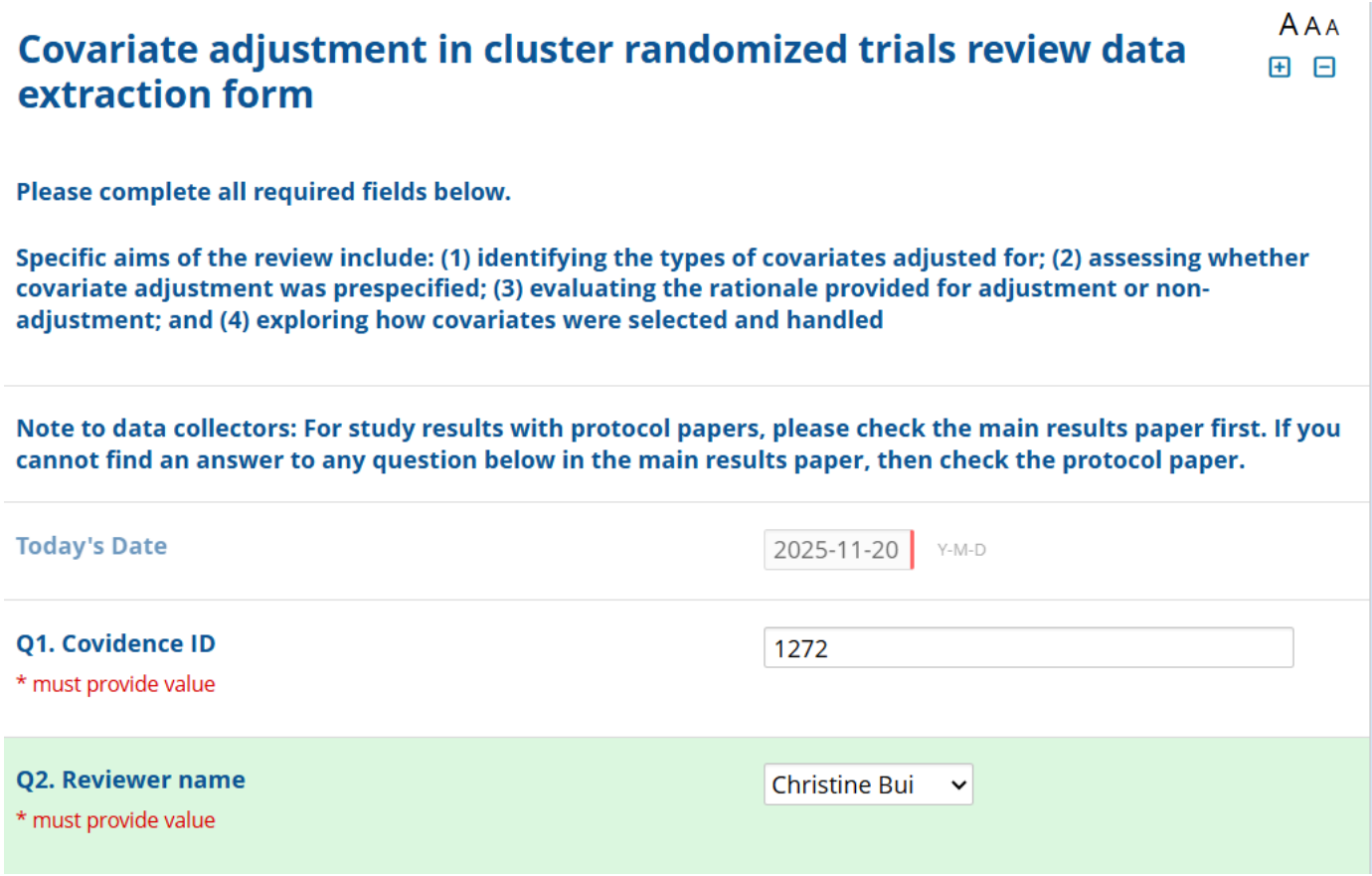
Data source

- Using 300 papers from 2017-2022
- All are parallel-arm cluster randomized trials conducted in LMICs
- Based on:
  - Ethics review of 800 cRCTs
  - Randomization procedures review using a random subset of 300

Shaw et al. 2025
Goldstein et al. 2024

Data extraction form

- REDCap
- Standard questions:
  - Total # of participants enrolled in the trial
  - Type of study
  - Primary outcome(s)
- Analysis questions:
  - How covariates were selected
  - Distributional summaries of covariates



Data extraction form (cont.)

Covariate adjustment

Q31. Copy and paste any text related to covariate adjustment, including rationale

* must provide value

Expand

Enter -999 if not reported or unclear

Q32. Were any covariates (other than baseline _____) used in the primary analysis?

* must provide value

- (•) Yes
- ( ) No
- ( ) Not reported or unclear

reset

Q33. Number of covariates (other than baseline _____) used in the primary analysis

* must provide value

Enter -999 if not reported or unclear

Q34. List of covariates used in the primary analysis

* must provide value

Expand

Names of variables, separated by semicolons

Q35. What types of covariates were used in the analysis?

* must provide value

- [ ] Cluster-level
- [ ] Individual-level
- [ ] Cluster Location
- [ ] Cluster Size
- [ ] Other
- [ ] Not reported or unclear

Covariate adjustment

Q31. Copy and paste any text related to covariate adjustment, including rationale

* must provide value

Expand

Enter -999 if not reported or unclear

Q32. Were any covariates (other than baseline _____) used in the primary analysis?

* must provide value

- ( ) Yes
- (•) No
- ( ) Not reported or unclear

reset

Q42. Specify rationale given by authors for not adjusting for covariates

* must provide value

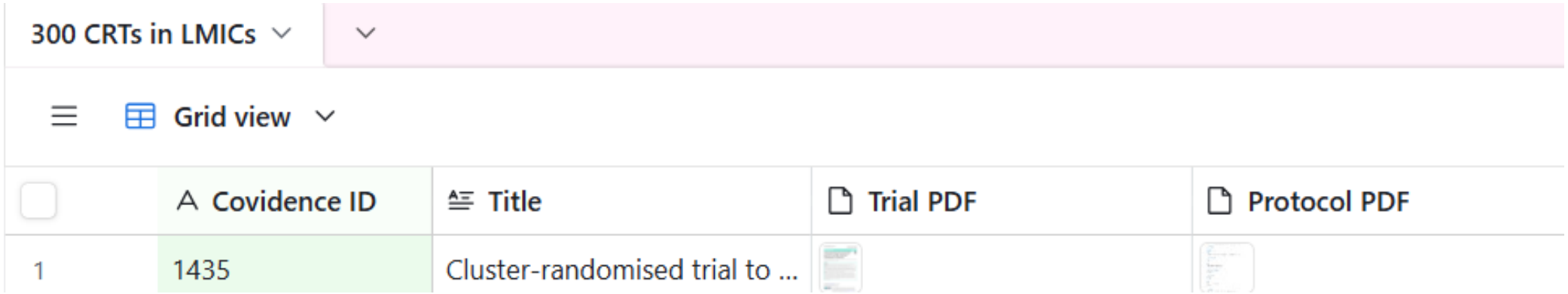
Enter -999 if not reported or unclear

Data extraction process

- Two reviewers per article
  - Senior reviewer and junior reviewer
- Random assignment of reviewer pairs and articles
- After individual article data extraction:
  - Reconciliation meeting
  - Resolve discrepancies
  - Document final decisions

Data extraction process (cont.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	week	Reviewer	Reviewer2	covidence	covidence	covidence	covidence	covidence	covidence	covidence	covidence	covidence	covidenceid10	
2	2	Alyssa	David	1760	1421	2173	2039	713	1550	265	1375	3197	3260	
3	2	Avi	Shifeng	1535	2288	2985	2893	2919	2643	2277	192	925	3099	
4	2	Joao	Zada	2020	2636	2218	3077	1173	2356	1378	1374	3330	2298	
5	2	Joe	Christine	1272	2070	934	558	2299	3291	3289	3281	2879	341	
6	2	John	Emma	1240	2480	2107	844	1703	908	1632	2072	2152	2702	
7	2	Liz	Qi	3004	922	3172	2627	1981	2644	1783	1532	1811	31	



Results & Next Steps

Results

- Article data extraction and reconciliation ongoing

Covariate adjustment

	Total (N = 182)
Q32. Were any covariates (other than baseline outcome) used in the primary analysis?	
Not reported or unclear	4 (2.2%)
No	70 (38.5%)
Yes	108 (59.3%)

Next steps

- Finish article data extraction and reconciliation
- Merge and clean data
- Create summary tables
- Identify patterns:
  - Which covariates are most common?
  - How often do authors give a rationale?
- Draft manuscript and think about how to frame main findings

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Thank you! Questions?

Special Thanks: John Gallis